

Wireless

CCNA 200-301 V1.1

MODULE 7

Topics

7.1 Wi-Fi Principles

7.2 Cisco Wireless Architectures

7.3 WLC Configuration

7.4 Review



7.1 Wi-Fi Principles

CCNA 200-301 v1.1

Module 7

How Wi-Fi Works

2.4 GHz, 5 GHz, 6 GHz Wi-Fi Channels

Wi-Fi 4, 5, 6, 7

Wi-Fi Security Levels

Wi-Fi Service Sets

How Wi-Fi Works

A wireless access point (WAP) is connected to the rest of the network

The WAP bridges the wireless and wired networks together

Wireless client devices (laptops, phones, IoT devices, etc.) connect to the WAP to access the network

In SOHO environments, the WAP is a built-in feature of the router

In enterprise environments, WAPs are separate devices

WAPs can either be managed individually as standalone devices, or centrally by a Wireless LAN Controller (WLC)



Wi-Fi Radio Channels

Wi-Fi uses radio channels in the 2.4 GHz, 5 GHz, and 6 GHz bands

A channel is actually a range frequencies

Data is spread across the channel range

In the 2.4 GHz band, most channels slightly overlap

Starting with 5 GHz, channels can be “ganged up”

- Creates “wider” channels
- Offers clients more bandwidth



Wi-Fi 4

802.11n

- Part of the unlicensed Industrial, Scientific, Medical (ISM) band
- 2.4 GHz, 5 GHz
- Up to 600 mb/s

14 overlapping channels

- Channels 1, 6, 11, and 14 are the only non-overlapping channels*
- Ch. 1 – 11 used in the US
- Ch. 12 – 13 also used by Europe
- Ch. 12 - 14 also used by Japan

2.4 GHz

- Able to reach farther/provide more coverage
- Crowded - many devices use 2.4 GHz frequency band, which causes interference
- Microwaves, cordless phones, baby monitors

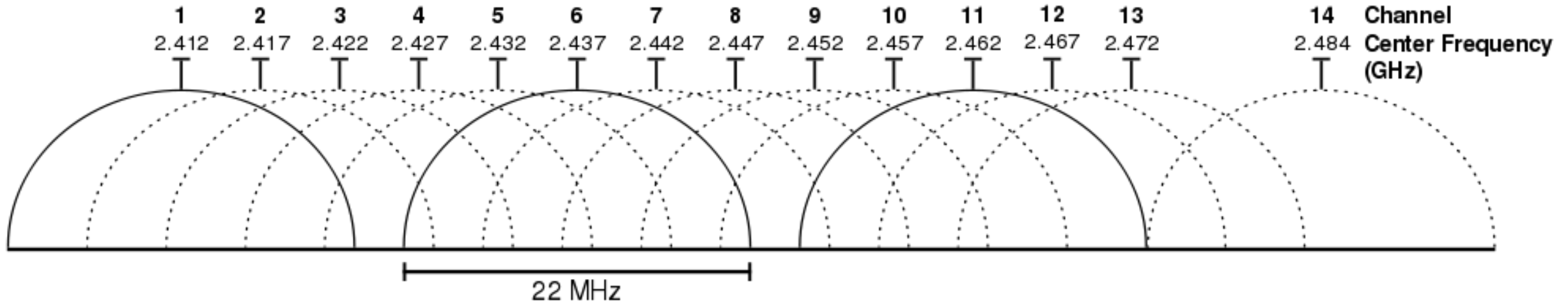
5 GHz

- Shorter antenna / shorter distance than 2.4 GHz
- Fewer interference sources than 2.4 GHz
- Potential interference by weather stations and military radar

*US only - other countries have different patterns



2.4 GHz and 5 GHz Wi-Fi Channels



2.4 GHz (802.11b/g/n)



Non-overlapping

5 GHz (802.11a/n/ac)



Wi-Fi 5

802.11ac

- 2.4 GHz, 5 GHz
- Up to 3.5 gb/s
- 5 GHz is also part of the ISM band
 - Weather stations and military radar share part of the band
 - Might cause interference

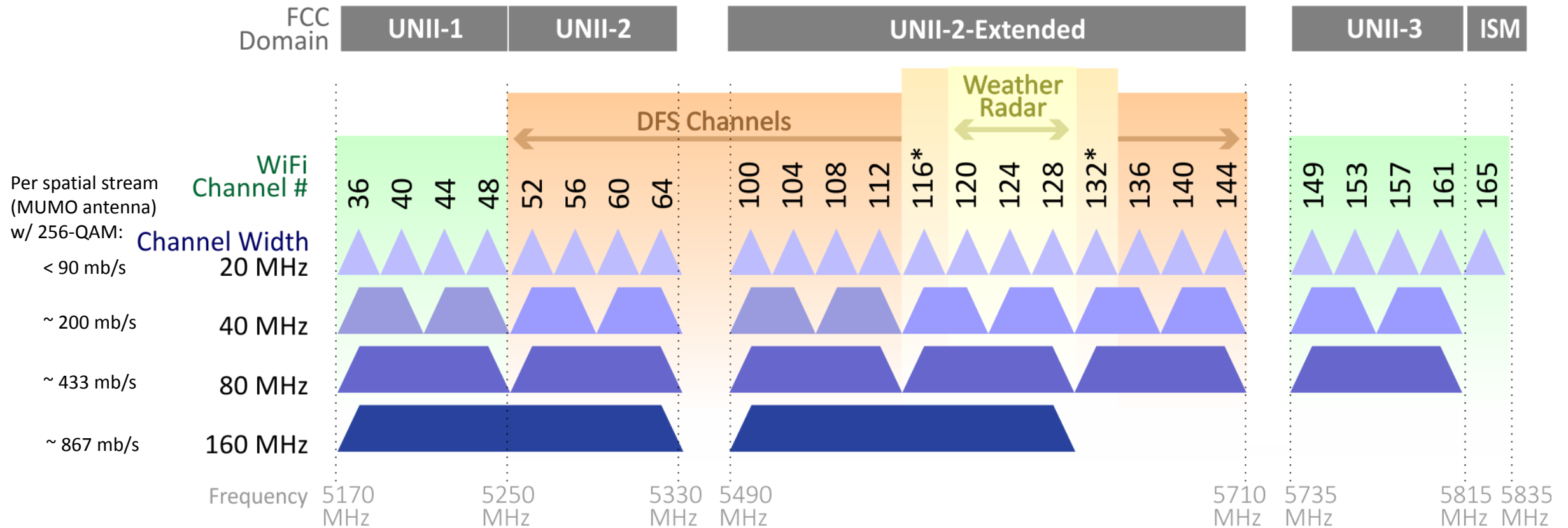
Up to 25 channels (depending on channel width)

No channel overlap

- 2.4 GHz channels 1, 6, and 11 only



Wi-Fi 5 Channel Map



*Channels 116 and 132 are Doppler Radar channels that may be used in some cases.

Credit: securityuncorked.com



Wi-Fi 6

802.11ax

2.4 GHz, 5 GHz

Wi-Fi 6E introduced 6 GHz

Up to 87 channels (depending on channel width)

No channel overlap

Up to 9.6 gb/s data rate

- 160 MHz channel x 8 spatial streams, 1024-QAM



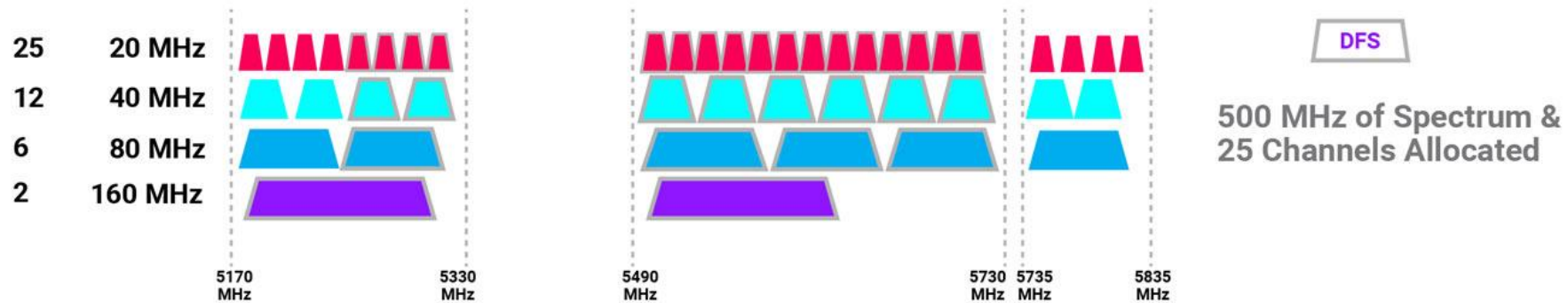
Wi-Fi 6E Channel Map

Band **Channels** **BW**

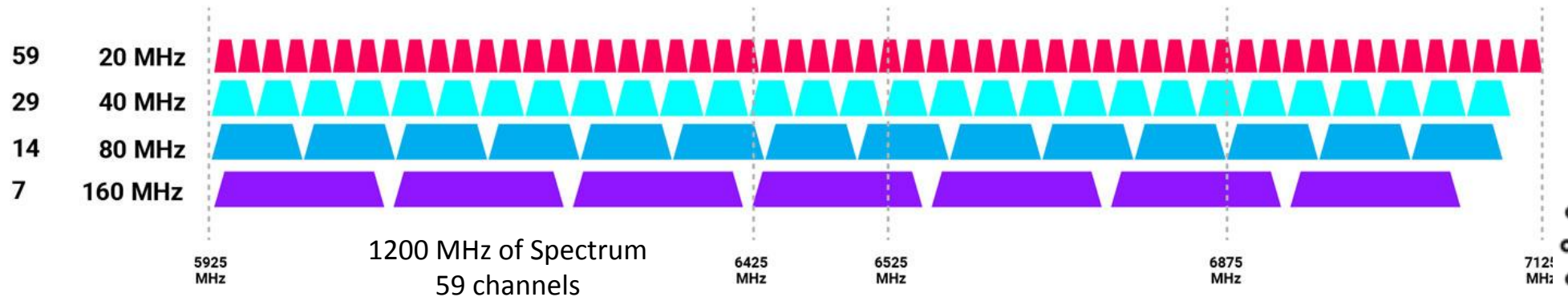
2.4 GHz

3	20 MHz		60 MHz of Spectrum & 3 Channels Allocated
1	40 MHz		

5 GHz



6 GHz



Wi-Fi 7

802.11be

2.4 GHz, 5 GHz, 6 GHz

Up to 46 gb/s max data rate

- 320 MHz channel width, 4096-QAM
- 16 spatial streams

Up to 116 non-overlapping channels

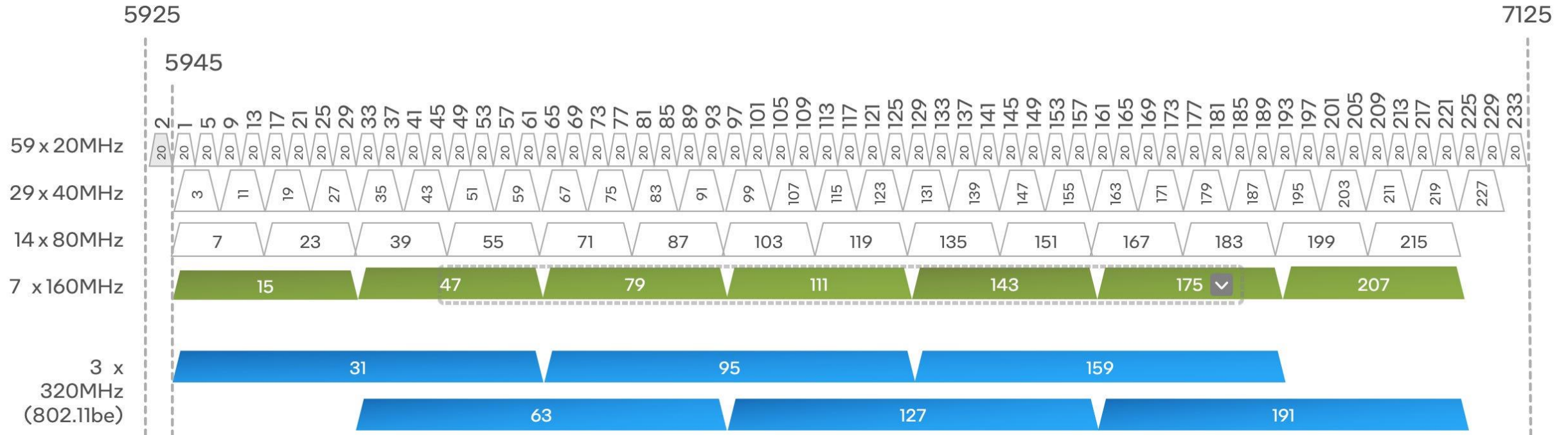
Driven by virtual reality (VR) applications, 8k video, large-scale IoT and gaming



ASUS GT-BE98 WiFi 7 Gaming Router



Wi-Fi 7 Channel Map



Wi-Fi Security Levels



Wi-Fi Security

Wi-Fi has two levels of security:

Personal:

- Configure a standalone WAP with a pre-shared key
- The user must enter the key when connecting to the access point

Enterprise:

- WAPs act as 802.1x clients
- When end devices (supplicants) connect, their connection is put on hold
- The user typically sees a captive portal where they must enter their credentials
- The WAP (RADIUS client) forwards the authentication request to a RADIUS server
- The RADIUS server informs the RADIUS client if the authentication request was successful, and if there are any policy restrictions to be applied
- If authentication is successful, the supplicant is allowed on the network



Comparison of WPA, WPA2, and WPA3

Feature	WPA	WPA2	WPA3
Encryption	TKIP	AES (CCMP) 128-bit key	AES (CCMP) 192-bit key
Authentication	Pre-shared key (PSK) Enterprise (802.1x + RADIUS)	PSK, Enterprise	Simultaneous Authentication of Equals (SAE) replaces PSK, Enterprise
Key Management	TKIP – each packet has a new key	Robust Security Network (AES CCMP) key management	Forward Secrecy New key every session
Protection Against	Weak WEP keys, anti-replay packet sequencing	TKIP vulnerabilities, stronger	Brute-force attacks, offline dictionary attacks
Introduced In	2003	2004	2018
Use Case	Legacy devices	Commonly used today	High security environment



Wi-Fi Service Sets



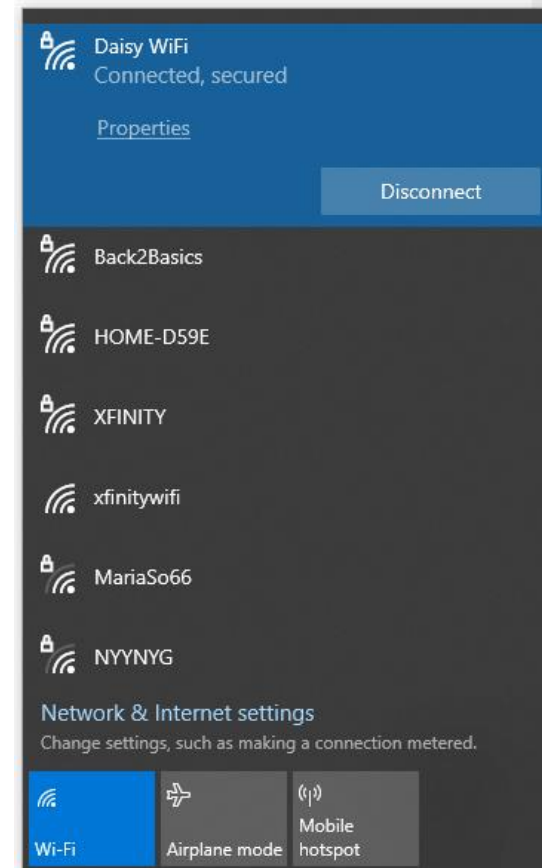
Service Set Identifier (SSID)

The friendly name given to a wireless network

Need not be unique

Can be hidden (not advertised)

- You can still connect to the WLAN if you know the SSID
- You'll have to manually enter the SSID



Basic Service Set (BSS)

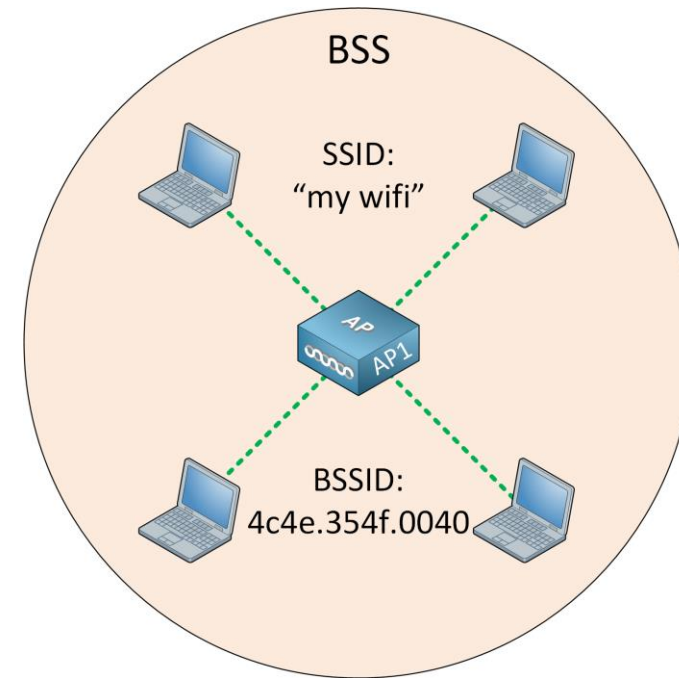
Simple WLAN with ONE:

- Wireless access point
- SSID (AP advertises itself)
- Channel
- BSSID (MAC address of AP)

Typically can accommodate up to 10 clients

Usually an extension of the LAN

Traffic might also be routed straight to the Internet



Extended Service Set (ESS)

Several interconnected BSSs acting as one

APs that are physically close to each other will use different channels

- Avoid interfering with each other

All participating BSSs use the same SSID

- To the client, the ESS appears as a single BSS

Depending on the product, the ESS might provide several SSIDs

- Each WAP in the ESS can accept connections to any or all of the SSIDs
- To the client, each SSID appears to be a different system, with its own security and network settings

APs in an ESS are typically managed by a separate WLAN controller

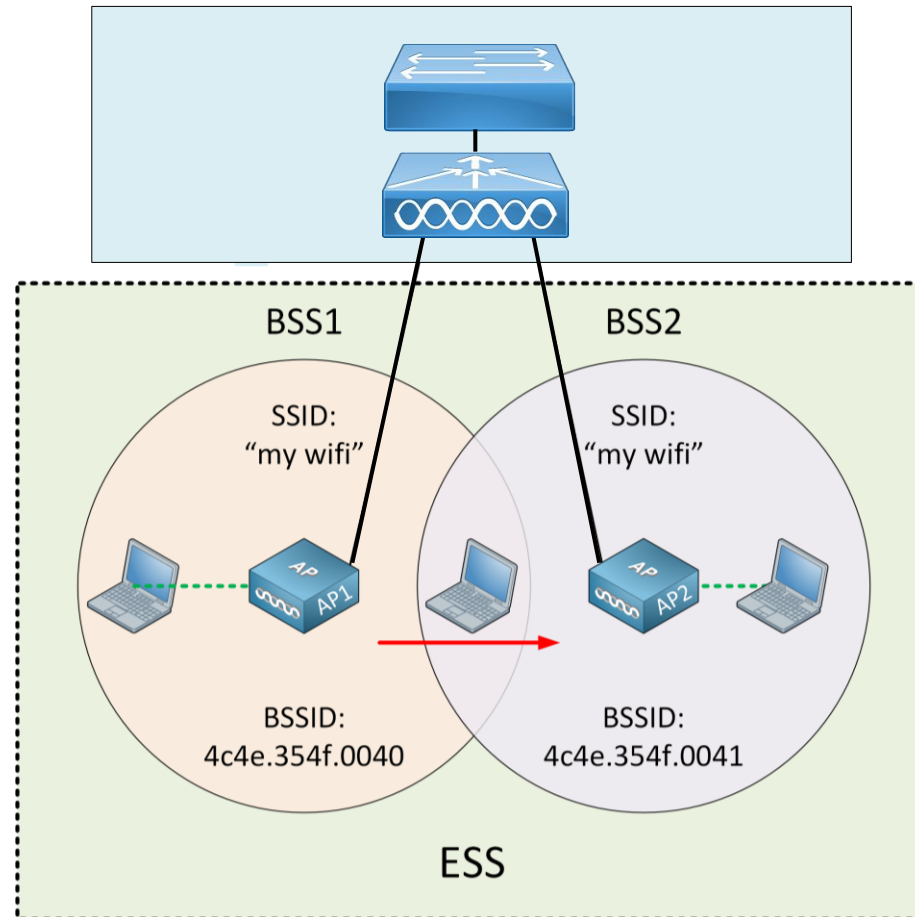
- The controller sends configuration information, including load balancing user traffic, to the APs
- Controller traffic and user data is tunneled between the controller and AP(s)

Can have thousands of active clients

- Depending on the product, each AP can handle 50 – 500+ concurrent clients per radio



ESS Example



7.2 Cisco Wireless Architectures

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Module 7

Wireless Access Point (WAP)

Wireless LAN Controller (WLC)

Cisco Lightweight Access Point (LAP) Connections

WLC – LAP Communications

Trunk Port Connections

Link Aggregation Group (LAG)

Centralized (Unified) Architecture

Distributed (FlexConnect) Architecture

Cloud-Based Architecture

Cisco Access Point Modes

Wireless Access Point (WAP)

AKA Access Point, AP

Can be single, dual, or tri-band

Can support 10 – 500+ simultaneous Wi-Fi clients

Can be standalone or WLC-controlled

- Cisco Lightweight Access Point (LAP) is WLC-controlled

Can be PoE or separately powered

Has at least one RJ-45 Ethernet port

Might have its own console port



Wireless LAN Controller (WLC)

Monitors and manages wireless access points

Connects the WAPs to the wired network

Communicates with each WAP via a dual CAPWAP tunnel

- One tunnel for control
- One tunnel for data



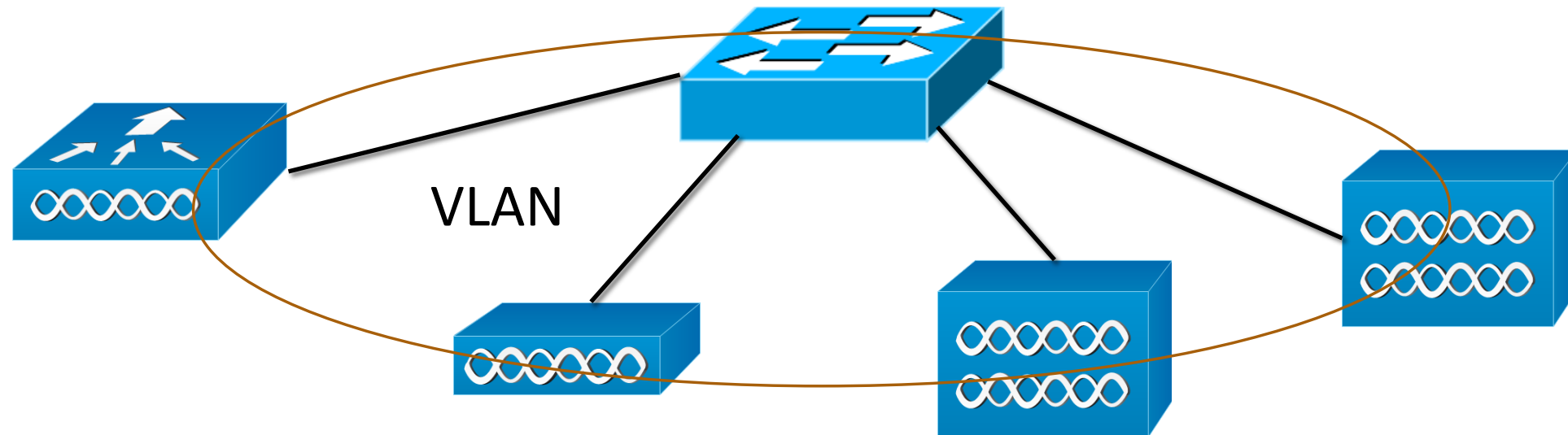
Cisco Lightweight Access Point (LAP) Connections

Cisco LAPs connect to access switch ports

These ports are placed in a dedicated LAP management VLAN

The WLC connects to the same VLAN to manage the LAPs and receive their Wi-Fi client traffic

- The WLC can be part of the switch, or a separate device



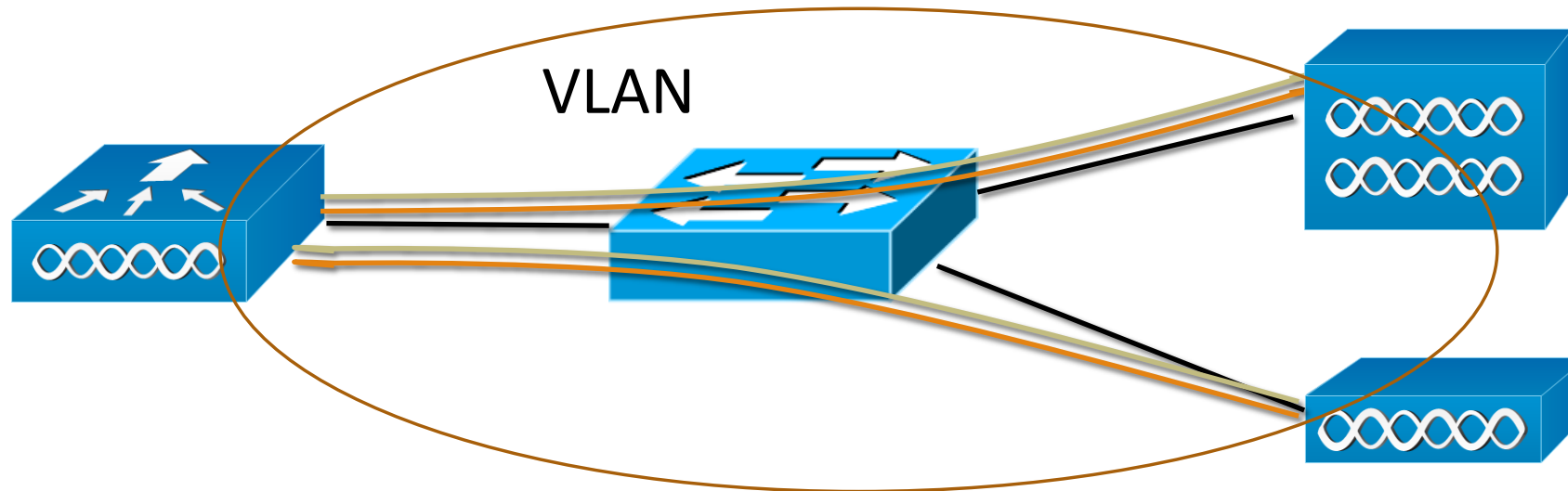
WLC – LAP Communications

Each LAP creates a Control and Provisioning of Wireless Access Points (CAPWAP) tunnel to the WLC

- CAPWAP is an IETF standard that replaced Cisco's proprietary Lightweight Access Point Protocol (LWAPP)

Each CAPWAP tunnel is actually two encrypted tunnels, one for control and one for data

All CAPWAP tunnels travel across the same VLAN to the WLC



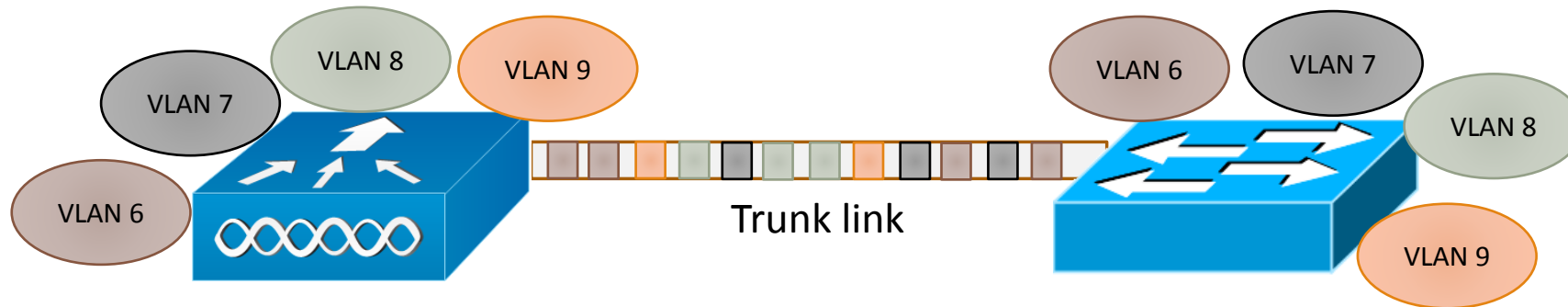
Trunk Port Connections

Used by a WLC to connect to its local access switch

- Needed because the WLC routes LAP traffic from the CAPWAP tunnels to the various VLANs

Can also be used by autonomous WAPs configured with multiple SSIDs

- Each SSID is assigned to its own VLAN
- The WAP uses a trunk link to send and receive all VLAN traffic to/from the switch



Link Aggregation Group (LAG)

Used to bundle multiple trunk links between a WLC and a switch

- Increases Wi-Fi client traffic throughput

Can be automatically configured using either LACP or PAgp

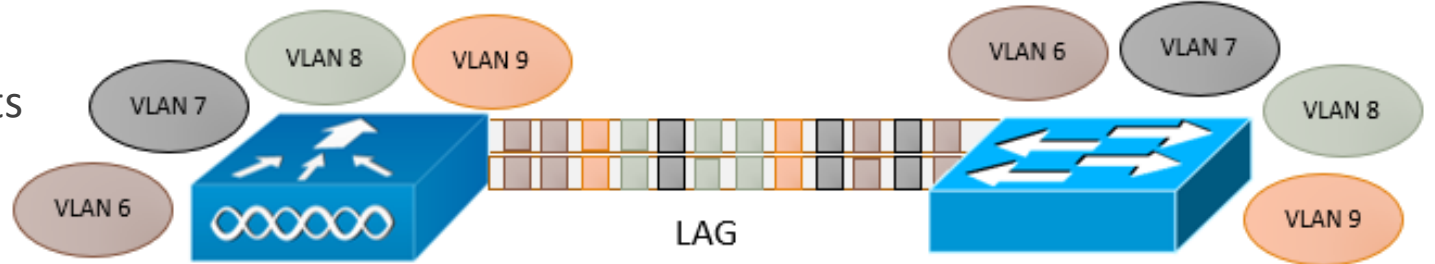
Link Aggregation Protocol (LACP)

- Standards-based, vendor neutral
- Good for multi-vendor environments
- Preferred in most implementations

Port Aggregation Protocol (PAgp)

- Cisco proprietary

Note: Cisco's term for LAG is "EtherChannel"



Centralized (Unified) Architecture

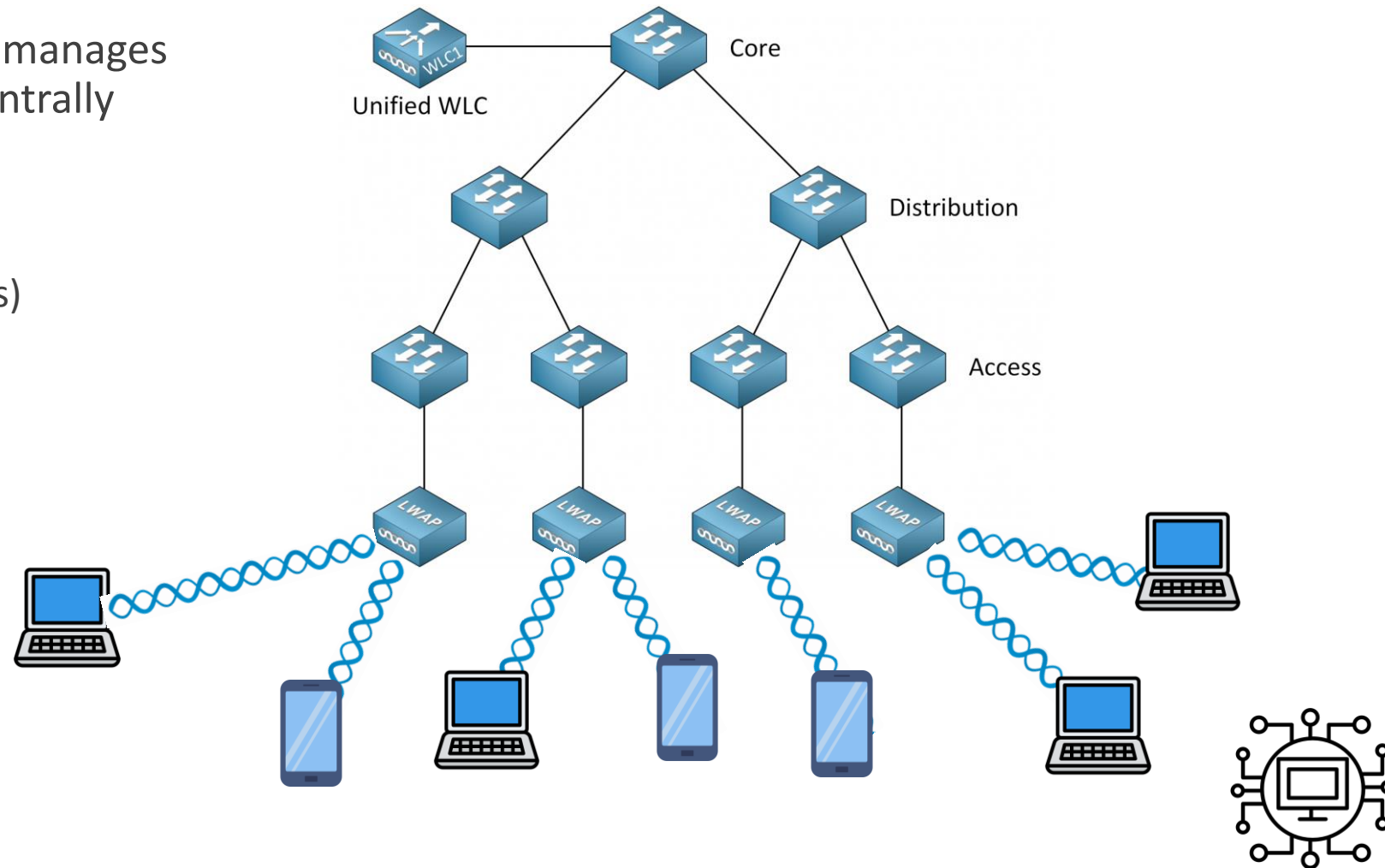
Wireless LAN Controller (WLC) manages multiple access points (APs) centrally

Components:

- Wireless LAN Controller (WLC)
- Lightweight Access Points (LAPs)

Benefits:

- Simplified management
- Consistent security policies
- Improved scalability



Distributed (FlexConnect) Architecture

For branch office and remote office deployments

Enables remote configuration and control of access points

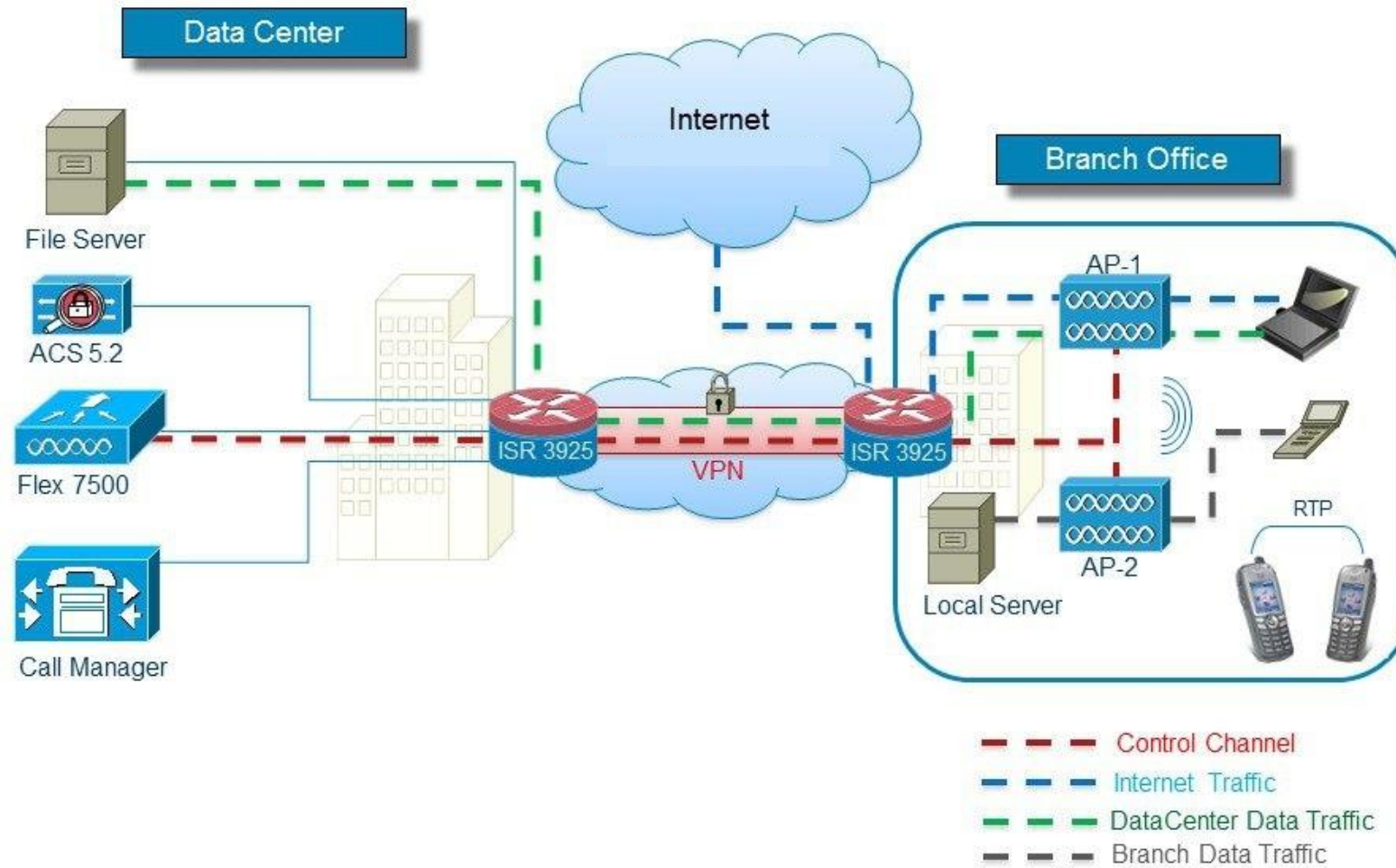
- Corporate office WLC → WAN → branch office AP
- Branch / remote offices do not have their own WLC

FlexConnect APs can:

- Switch client data traffic locally
- Perform client authentication locally
- When connected to the controller, also send traffic back to the controller



FlexConnect Architecture Example



Cloud-Based Architecture

Uses cloud management platforms like Cisco Meraki for centralized management via the cloud

- WLC is in the cloud, not on premises

Components:

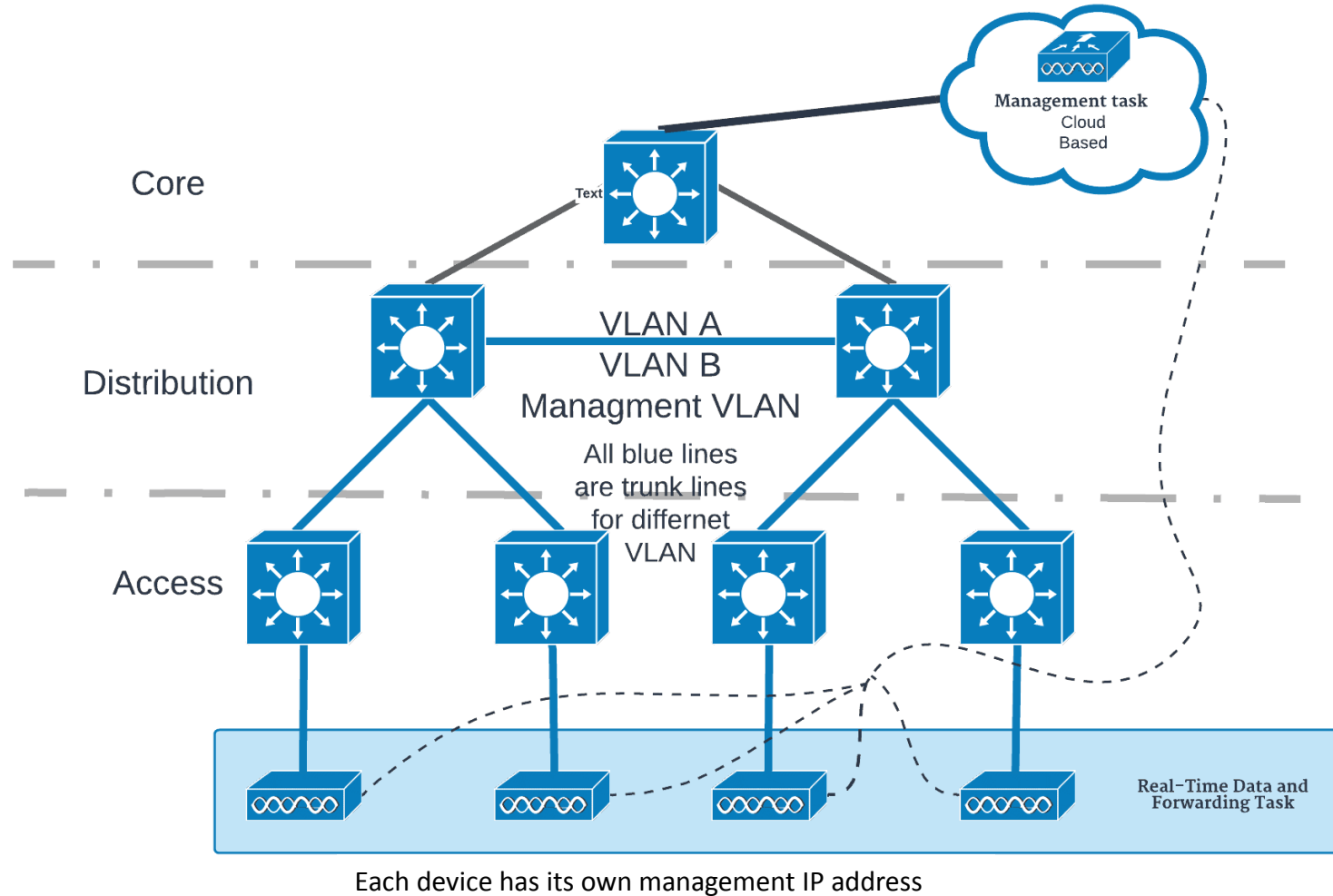
- Cloud Controller (Control Plane)
- On-premises APs (Data Plane)
- APs connect to the cloud via HTTPS for management traffic
- Wi-Fi client traffic stays on-premises

Benefits:

- Simplified deployment and management
- Scalability
- Remote access and monitoring



Cloud-Based Architecture Example



Cisco Access Point Modes

Typically configured on the WLC

Mode	Description
Local	Default mode for lightweight APs (LAPs) The LAPs serve clients and communicate with the WLC Ideal for environments with reliable WLC connections
FlexConnect	APs can switch traffic locally and perform client authentication when the WLC is unreachable For remote sites with limited WAN bandwidth or unreliable WLC connections
Bridge	APs can form a wireless bridge, connecting different wired networks Used for point-to-point or point-to-multipoint connections
Flex+Bridge	Combines FlexConnect and Bridge modes For complex environments requiring both local switching and bridging capabilities



Cisco Access Point Modes (cont'd)

Mode	Description
Sniffer	APs act as packet sniffers, capturing wireless traffic for analysis Useful for troubleshooting and performance monitoring
Monitor	APs scan all channels for rogue devices and security threats Enhances network security by detecting unauthorized devices
Rogue Detector	APs monitor the wired network for rogue devices Helps in identifying unauthorized APs connected to the wired network
SE-Connect	APs provide spectrum analysis to identify sources of interference Useful for RF planning and troubleshooting

7.3 WLC Configuration

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Module 7

Common Configurable WLC Features

WLC GUI Configuration Examples

Fast Transition (802.11r)

Common Configurable WLC Features

New WLANs

- SSIDs, VLANs, IP addresses, WPA/WPA2 encryption, Pre-shared Key, AAA and 802.1x authentication

Quality of Service (QoS) Profiles

- Platinum (Voice)
- Gold (Video)
- Silver (Best Effort)
- Bronze (Background)

Data Rate support

- You can specify the data rates that clients MUST be able to support to connect (from 1 mbps – 9.6 gbps)
- You typically want to disable the “legacy” rates of 1mbps, 2mbps, 5.5mbps and 11mbps to improve network efficiency



Common Configurable WLC Features (cont'd)

RF Band support

- 2.4 GHz, 5 GHz, 6 GHz
- Channels and channel width

Access Point Mode

- Local, FlexConnect, Bridge, Flex+Bridge, Sniffer, Monitor, Rogue Detector, SE-Connect

Client Band Select

- Attempt to aggressively steer dual-band clients to the less congested 5 GHz band

Peer-to-peer blocking

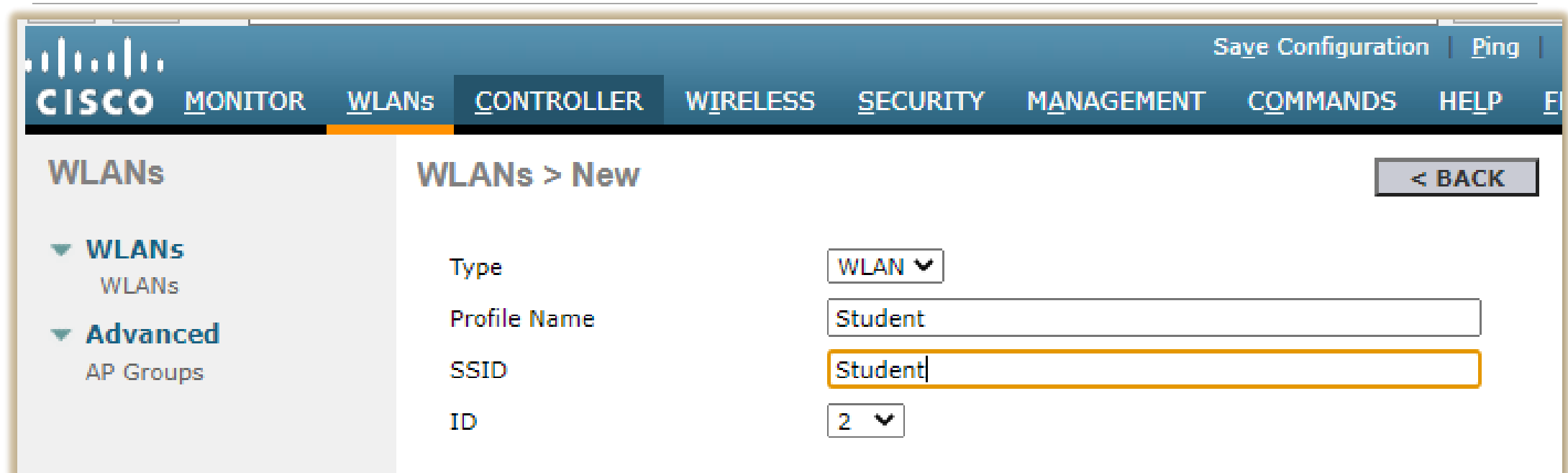
- Wi-Fi clients cannot use the WAP to connect to each other

Rogue detection and containment

- Over-the-air or connected to the LAN



WLC - WLAN Creation



The screenshot shows the Cisco Wireless LAN Controller (WLC) web interface. The top navigation bar includes links for MONITOR, WLANs, CONTROLLER, WIRELESS, SECURITY, MANAGEMENT, COMMANDS, and HELP. The 'WLANs' tab is selected. On the left, a sidebar shows 'WLANs' and 'Advanced' sections. The main content area is titled 'WLANs > New' and contains a form for creating a new WLAN. The form fields are: Type (WLAN), Profile Name (Student), SSID (Student), and ID (2). A '< BACK' button is located in the top right corner of the form area.

Save Configuration | Ping |

CISCO MONITOR **WLANs** CONTROLLER WIRELESS SECURITY MANAGEMENT COMMANDS HELP

WLANs

WLANs > New < BACK

Type WLAN

Profile Name Student

SSID Student

ID 2



WLC – Manage Access Points



[MONITOR](#)[WLANs](#)[CONTROLLER](#)[WIRELESS](#)[SECURITY](#)[MANAGEMENT](#)

Wireless

▶ Access Points

▶ Advanced

Mesh

▶ ATF

RF Profiles

FlexConnect Groups

FlexConnect ACLs

FlexConnect VLAN

Templates

QoS ACLs

All APs

Current Filter

[\[Change Filter\]](#) [\[Clear Filter\]](#)

Number of APs 3

AP Name	IP Address(Ipv4/Ipv6)
Light Weight Access Point1	192.168.1.11
Light Weight Access Point0	192.168.1.10
Light Weight Access Point2	192.168.1.12



WLC - Configure WLAN WPA2 PSK



WLANs

WLANs > Edit 'Student'

< BACK Apply

General Security QoS Policy-Mapping Advanced

Layer 2 Layer 3 AAA Servers

Layer 2 Security **WPA+WPA2** ☐ MAC Filtering ☐

Fast Transition ☐

Protected Management Frame

PMF Disabled

WPA+WPA2 Parameters

WPA Policy ☐

WPA2 Policy ☒

WPA2 Encryption ☒ AES ☐ TKIP

Authentication Key Management

802.1X ☐ Enable

CCKM ☐ Enable

PSK ☒ Enable

WLANs

WLANs > Edit 'Student'

<

General Security QoS Policy-Mapping Advanced

Layer 2 Layer 3 AAA Servers

Layer 2 Security **WPA+WPA2** ☐ MAC Filtering ☐

Protected Management Frame

PMF Disabled

WPA+WPA2 Parameters

WPA Policy ☐

WPA2 Policy ☒

WPA2 Encryption ☒ AES ☐ TKIP

Authentication Key Management

802.1X ☐ Enable

CCKM ☐ Enable

PSK ☒ Enable

FT 802.1X ☐ Enable

FT PSK ☐ Enable

PSK Format ASCII

.....

WPA+WPA2 Parameters

WPA Policy ☐

WPA2 Policy ☒

WPA2 Encryption ☒ AES ☐ TKIP

Authentication Key Management

802.1X ☐ Enable

CCKM ☐ Enable

PSK ☒ Enable

FT 802.1X ☐ Enable

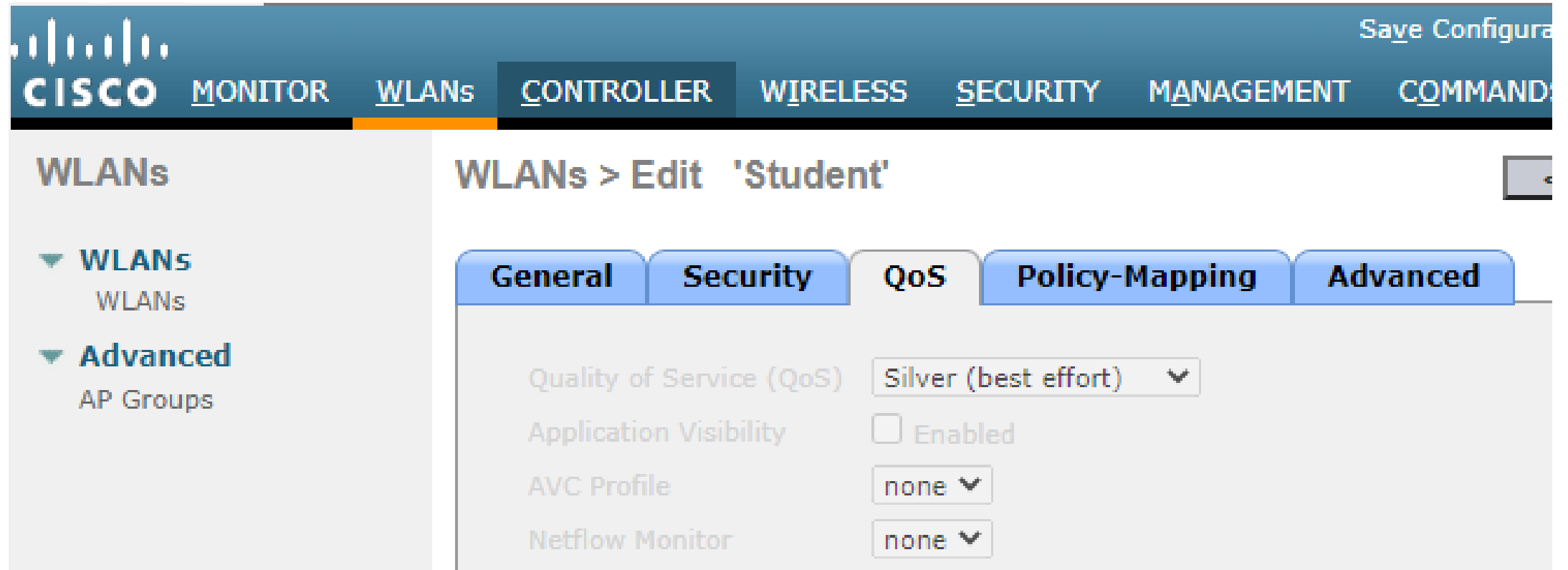
FT PSK ☐ Enable

PSK Format ASCII

.....



WLC - Configure WLAN QoS



The screenshot shows the Cisco Wireless LAN Controller (WLC) configuration interface. The top navigation bar includes the Cisco logo and tabs for MONITOR, WLANs, CONTROLLER, WIRELESS, SECURITY, MANAGEMENT, and COMMANDS. The 'WLANs' tab is selected. On the left, a sidebar shows 'WLANs' and 'Advanced' (AP Groups) sections. The main content area is titled 'WLANs > Edit 'Student'' and features five tabs: General, Security, QoS, Policy-Mapping, and Advanced. The 'QoS' tab is active, displaying configuration options for Quality of Service (QoS), Application Visibility, AVC Profile, and Netflow Monitor.

WLANs

- ▼ **WLANs**
WLANs
- ▼ **Advanced**
AP Groups

WLANs > Edit 'Student'

General **Security** **QoS** **Policy-Mapping** **Advanced**

Quality of Service (QoS) Silver (best effort) ▼

Application Visibility ☐ Enabled

AVC Profile none ▼

Netflow Monitor none ▼



WLC - Add a AAA Server

CISCO

MONITOR

WLANs

CONTROLLER

WIRELESS

SECURITY

MANAGEMENT

COMMANDS

HELP

FEEDBACK

Save Configuration

Ping

Logout

Refresh

Security

AAA

General

RADIUS

Authentication

Accounting

Fallback

DNS

Downloaded AVP

TACACS+

LDAP

Local Net Users

MAC Filtering

Disabled Clients

User Login Policies

AP Policies

Downloaded Policies

RADIUS Authentication Servers

Auth Called Station ID TypeIP Address

Use AES Key Wrap☐ (Designed for FIPS customers and requires a key wrap compliant RADIUS server)

MAC DelimiterHyphen

Framed MTU1300

Network User	Management	Server Index	Server Address(Ipv4/Ipv6)	Port
☒	☒	1	192.168.1.253	1812



Configure the WLAN to Use the AAA Server

WLANs

- ▼ **WLANs**
WLANs
- ▼ **Advanced**
AP Groups

WLANs > Edit 'Student'

General **Security** **QoS** **Policy-Mapping**

Layer 2 **Layer 3** **AAA Servers**

Select AAA servers below to override use of default server

Radius Servers

Radius Server Overwrite interface ☐ Enabled

	Authentication Servers	Accounting
	☒ Enabled	☐ Enabled
Server 1	IP:192.168.1.253, Port:1812 ▼	None ▼
Server 2	None ▼	None ▼



WLC - WLAN Advanced Settings

Allow the
RADIUS
server to
set the
VLAN

The screenshot shows the 'Advanced' tab of the WLC configuration page. The 'General' sub-tab is active. The 'Allow AAA Override' checkbox is checked and highlighted with a red box. The 'Client Exclusion' checkbox is also checked, with a timeout value of 180 seconds. The 'Maximum Allowed Clients' is set to 0. The 'Static IP Tunneling' checkbox is unchecked. The 'Wi-Fi Direct Clients Policy' is set to 'Disabled'. The 'Maximum Allowed Clients Per AP Radio' is set to 200. The 'Clear HotSpot Configuration' checkbox is unchecked. The 'DHCP' section shows 'DHCP Server' and 'DHCP Addr. Assignment' both unchecked. The 'Management Frame Protection (MFP)' section shows 'MFP Client Protection' set to 'Optional'. The 'DTIM Period (in beacon intervals)' section shows '802.11a/n (1 - 255)' and '802.11b/g/n (1 - 255)' both set to 1. The 'NAC' section shows 'NAC State' set to 'None'. The 'Load Balancing and Band Select' section shows 'Client Load Balancing' and 'Client Band Select' both unchecked, with the latter highlighted by a red box. The 'Passive Client' section is partially visible.

Section	Setting	Value
General	Allow AAA Override	☒ Enabled
	Coverage Hole Detection	☐ Enabled
	Enable Session Timeout	☐
	Aironet IE	☐ Enabled
	Diagnostic Channel	☐ Enabled
	Override Interface ACL	IPv4: None, IPv6: None
	Layer2 Acl	None
	URL ACL	None
	P2P Blocking Action	Disabled
	Client Exclusion	☒ Enabled, 180 Timeout Value (secs)
	Maximum Allowed Clients	0
	Static IP Tunneling	☐ Enabled
	Wi-Fi Direct Clients Policy	Disabled
	Maximum Allowed Clients Per AP Radio	200
	Clear HotSpot Configuration	☐ Enabled
DHCP	DHCP Server	☐ Override
	DHCP Addr. Assignment	☐ Required
	Management Frame Protection (MFP)	
	MFP Client Protection	Optional
DTIM Period (in beacon intervals)	802.11a/n (1 - 255)	1
	802.11b/g/n (1 - 255)	1
NAC	NAC State	None
Load Balancing and Band Select	Client Load Balancing	☐
	Client Band Select	☐
Passive Client		

Redirect dual band
clients to 5 GHz



Rogue Detection and Containment

CISCO

MONITOR WLANS CONTROLLER WIRELESS SECURITY MANAGEMENT COMMANDS HELP FEEDBACK

Save Configuration Ping Logout Refresh Home

Security

- AAA
 - General
 - RADIUS
 - Authentication
 - Accounting
 - Fallback
 - DNS
 - Downloaded AVP
 - TACACS+
 - LDAP
 - Local Net Users
 - MAC Filtering
 - Disabled Clients
 - User Login Policies
 - AP Policies
 - Password Policies
- Local EAP
- Advanced EAP
- Priority Order
- Certificate
- Access Control Lists
- Wireless Protection Policies
 - Rogue Policies
 - General
 - Rogue Rules
 - Friendly Rogue
 - Standard Signatures
 - Custom Signatures
 - Signature Events
 - Summary
 - Client Exclusion Policies
 - AP Authentication
 - Management Frame Protection

Rogue Policies

Apply

Rogue Detection Security Level

☐ Low ☐ High ☐ Critical ☒ Custom

Rogue Location Discovery Protocol

Expiration Timeout for Rogue AP and Rogue Client entries Seconds

Validate rogue clients against AAA ☐ Enabled

Validate rogue AP against AAA ☐ Enabled

Polling Interval Seconds

Validate rogue clients against MSE ☐ Enabled

Detect and report Ad-Hoc Networks ☒ Enabled

Rogue Detection Report Interval (10 to 300 Sec)

Rogue Detection Minimum RSSI (-70 to -128)

Rogue Detection Transient Interval (0, 120 to 1800 Sec)

Rogue Client Threshold (0 to disable, 1 to 256)

Rogue containment automatic rate selection ☐ Enabled

Auto Contain

Auto Containment Level

Auto Containment only for Monitor mode APs ☐ Enabled

Auto Containment on FlexConnect Standalone ☐ Enabled

Rogue on Wire ☐ Enabled

Using our SSID ☐ Enabled

Valid client on Rogue AP ☐ Enabled

AdHoc Rogue AP ☐ Enabled



Fast Transition (802.11r)

802.11r

Used to minimize disruption when a Wi-Fi client roams between APs in the same network

- Especially useful for real-time communications such as voice, video, collaboration
- Enhances the performance of real-time apps
- Reduces latency/handoff times - minimizes time required for client devices to roam between APs
- Seamless for users in environments with multiple APs
- Allows a client to pre-authenticate to multiple APs before the user roams

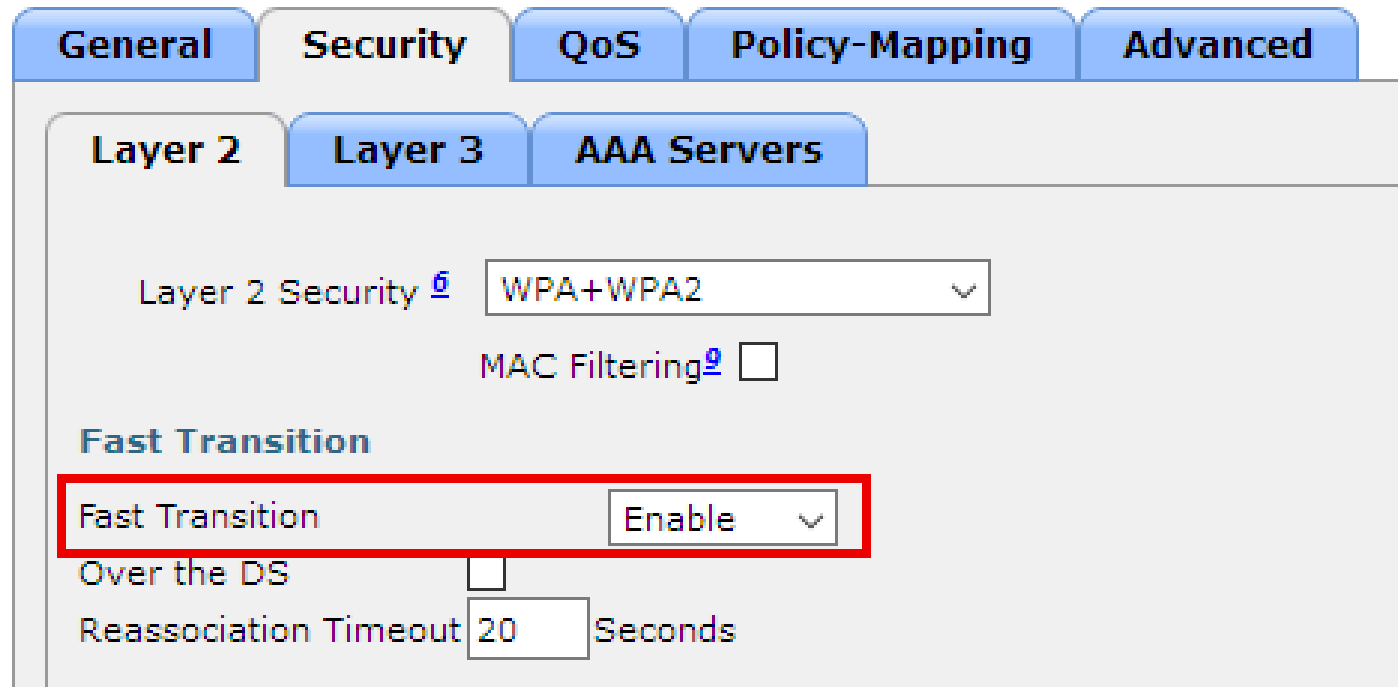
Can be implemented in two modes:

- Over-the-Air
 - The client directly communicates with the new AP it's about to roam to
- Over-the-DS (Distribution System)
 - The client communicates with the new AP via the WLC



Configuring Fast Transition on a WLC

WLANS → <Your WLAN> → Edit



The screenshot shows the configuration page for a WLAN on a Cisco Wireless LAN Controller (WLC). The page has a top navigation bar with tabs: General, Security, QoS, Policy-Mapping, and Advanced. Below this, there are sub-tabs: Layer 2, Layer 3, and AAA Servers. The Layer 2 tab is selected. Under the Layer 2 tab, there is a section for Layer 2 Security, which includes a dropdown menu for Layer 2 Security (set to WPA+WPA2) and a checkbox for MAC Filtering (unchecked). Below this is a section for Fast Transition, which includes a dropdown menu for Fast Transition (set to Enable) and a checkbox for Over the DS (unchecked). The Fast Transition dropdown is highlighted with a red box. Below the Fast Transition section, there is a field for Reassociation Timeout, set to 20 Seconds.

General Security QoS Policy-Mapping Advanced

Layer 2 Layer 3 AAA Servers

Layer 2 Security [6](#) WPA+WPA2

MAC Filtering [9](#) ☐

Fast Transition

Fast Transition Enable

Over the DS ☐

Reassociation Timeout 20 Seconds

Review

7.4 Review

CCNA 200-301 v1.1

Module 7

Review

Wireless access points (WAPs) act as a bridge between wireless and wired networks

- The WAP connects to the wired network
- Wireless client devices connect to the WAP

Wi-Fi uses radio channels in the 2.4 GHz, 5 GHz, and 6 GHz bands

- Wi-Fi 6E introduced the 6 GHz band

Each version of Wi-Fi increased throughput with:

- More channels
- Wider channels
- More spatial streams (radio-antenna pairs)
- Denser modulation

We are currently moving into Wi-Fi 7

- 46 gb/s max throughput
- Meant for 8k video, VR, large-scale IoT



Review (cont'd)

Wi-Fi security for SOHO environments uses a pre-shared key or SAE mutual authentication

WPA introduced TKIP key rotation to address the security weaknesses of WEP

WPA2 introduced AES CCMP 128-bit (much stronger) encryption

- Includes data integrity and authenticity checks

WPA3 introduced (Perfect) Forward Secrecy

- Uses the Simultaneous Authentication of Equals (SAE) handshake to replace pre-shared key
- Fresh key every session
- Key derived from Diffie-Hellman key exchange
- Even with the network password, an attacker cannot retroactively decrypt previous communications
- AES CCMP encryption increased to 192-bit

In enterprise environments, WAPs act as RADIUS clients

- They place the supplicant device “on hold” as they forward user credentials to a central RADIUS server



Review (cont'd)

A Service Set Identifier (SSID) is the friendly name given to a wireless network

A Basic Service Set (BSS) has a single standalone WAP with one SSID

- Can handle a limited number of clients

An Extended Service Set (ESS) has multiple interconnected BSSs acting as one

- They share the same SSID
- WAPs that are physically close to each other are on different channels
- WAPs are typically managed by a separate WLAN controller
- Depending on the product, each WAP can handle 50 – 500+ concurrent clients per radio



Review (cont'd)

WAPs can be single, dual, or tri-band

- May have radios and antennas for 2.4 GHz, 5 GHz, and 6 GHz

WAPs can:

- Be standalone or WLC-controlled
- Have their own power supply or be powered via its twisted pair Ethernet cable (PoE)
- Have their own console port

All WAPs have at least one RJ-45 Ethernet port

- Cisco Lightweight Access Point (LAP) is WLC-controlled



Review (cont'd)

A Wireless LAN Controller (WLC):

- Monitors and manages wireless access points
- Connects the WAPs to the wired network
- Communicates with each WAP via a dual CAPWAP tunnel
 - One tunnel for control
 - One tunnel for data

Cisco Lightweight Access Points (LAPs) connect to access switch ports

- They are placed in a dedicated LAP management VLAN
- The WLC connects to the same VLAN to manage the LAPs and receive their Wi-Fi client traffic
- The WLC can be part of the switch, or a separate device



Review (cont'd)

Autonomous WAPs and WLCs typically use trunk links to connect to the LAN switch

- This allows them to communicate with many VLANs at once
- Each SSID is usually placed in its own VLAN

Link Aggregation Groups (LAGs) are used to bundle several trunk links together to increase throughput

- Cisco's implementation of a LAG is also referred to as "EtherChannel"

LAGs can be automatically created using:

- Open standard LACP
- Or Cisco proprietary PAgp



Review (cont'd)

Cisco's Centralized (Unified) Architecture uses a WLC to manage multiple LAPs

Cisco's Distributed (FlexConnect) Architecture uses remote configuration and control of access points for branch and remote offices

- The WLC is at the central office
- The local FlexConnect WAPs can:
 - Switch client data traffic locally
 - Perform client authentication locally
 - When connected to the controller, also send traffic back to the controller

Cloud-based architecture uses cloud management platforms like Cisco Meraki for centralized management via the cloud

- WLC is in the cloud, not on premises
- Client traffic stays on-premises, though it can also be routed up into the cloud



Review (cont'd)

A Cisco LAP can be configured for different modes

- Local – Connect on the LAN to their WLC
- FlexConnect - APs can switch traffic locally and perform client authentication when the WLC is unreachable
- Bridge - APs can form a wireless bridge, connecting different wired networks with point-to-point or point-to-multipoint connections
- Flex+Bridge - Combines FlexConnect and Bridge Modes
- Sniffer – captures traffic for analysis
- Monitor - APs scan all channels for rogue devices and security threats
- Rogue Detector - APs monitor the wired network for connected rogue Aps
- SE-Connect - APs provide spectrum analysis to identify sources of interference

AP modes are typically configured on the WLC



Review (cont'd)

Common settings you can configure on the WLC include:

New WLANs

- SSIDs, VLANs, IP addresses, WPA/WPA2 encryption, Pre-shared Key, AAA and 802.1x authentication

Quality of Service (QoS) Profiles

- Platinum (Voice)
- Gold (Video)
- Silver (Best Effort)
- Bronze (Background)

Data Rate support

- You can specify the data rates that clients **MUST** be able to support to connect (from 1 mbps – 9.6 gbps)
- You typically want to disable the “legacy” rates of 1mbps, 2mbps, 5.5mbps and 11mbps to improve network efficiency



Review (cont'd)

(Continued) Common settings you can configure on the WLC include:

RF Band support

- 2.4 GHz, 5 GHz, 6 GHz
- Channels and channel width

Access Point Mode

- Local, FlexConnect, Bridge, Flex+Bridge, Sniffer, Monitor, Rogue Detector, SE-Connect

Client Band Select

- Attempt to aggressively steer dual-band clients to the less congested 5 GHz band

Peer-to-peer blocking

- Wi-Fi clients cannot use the WAP to connect to each other

Rogue detection and containment

- Over-the-air or connected to the LAN



Review (cont'd)

Fast Transition (802.11r) is used to minimize disruption when a Wi-Fi client roams between APs in the same network

Can be implemented in two modes:

- Over-the-Air
 - The client directly communicates with the new AP it's about to roam to
- Over-the-DS (Distribution System)
 - The client communicates with the new AP via the WLC